

YanJun Chen 陈彦筠

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Research Interests

Reinforcement Learning for Large Language Models and Embodied Agents: models are trainable; the environments that train them are not. Toward making the environment trainable, the way models are.

Education

The Hong Kong Polytechnic University, Hong Kong

Sept 2024 to present

PhD, Department of Computing. Candidature confirmed June 2026.

Advisors: Prof. Wenjie Li (Maggie), Prof. Wei Zhang. Joint doctoral training with the Eastern Institute of Technology (EIT), Ningbo.

Jinan University, Guangzhou, China

Sept 2020 to Jun 2024

B.Eng. in Cyberspace Security. GPA 88.6 / 100 · Rank 6 of 62 · Outstanding Graduate.

Experience

Microsoft Research Asia, Singapore

Sept 2026 to present

Research Intern.

Publications

First-Author

- YanJun Chen**, Jinghan Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Action Shaping: Policies Absorb What They Can Express.** *arXiv:2609.32752*, 2026 (in submission).
- YanJun Chen**, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **The Trace Is the State: Exact Credit Assignment for LLM Agent Teams.** *arXiv:2603.06859*, 2026 (in submission).
- YanJun Chen**, Yirong Sun, Xinghao Chen, Jian Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Integrating Chain-of-Thought for Multimodal Alignment: A Study on 3D Vision-Language Learning.** *arXiv:2503.06232*, 2025. [dataset]
- YanJun Chen**, Xinming Zhang, Xianghui Wang, Zhiqiang Xu, Xiaoyu Shen, Wei Zhang. **Rethinking Soft Actor-Critic in High-Dimensional Action Spaces: The Cost of Ignoring Distribution Shift.** *arXiv:2410.16739*, 2024.
- YanJun Chen**, Dawei Zhu, Yirong Sun, Xinghao Chen, Wei Zhang, Xiaoyu Shen. **The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models.** *EMNLP 2024*.

Co-Authored

- Jun Xue, Junze Wang, Shanze Wang, Xinming Zhang, **YanJun Chen**, Wei Zhang. **FastDSAC: Unlocking the Potential of Maximum Entropy RL in High-Dimensional Humanoid Control.** *NeurIPS 2026*.
- Xinghao Chen, Anhao Zhao, Heming Xia, Xuan Lu, Hanlin Wang, **YanJun Chen**, Wei Zhang, Jian Wang, et al. **Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning.** *Findings of EMNLP 2026*.
- Yuhan Hu, Yirong Sun, **YanJun Chen**, Xinghao Chen, Xiaoyu Shen, Wei Zhang. **Breaking the Pre-Planning Barrier: Adaptive Real-Time Coordination of Heterogeneous UAVs.** *IEEE Transactions on Vehicular Technology*, 2026.
- Jinghan Wang, Gaoliang Peng, **YanJun Chen**, Wei Zhang, Wentao Wu, Tianchen Liu. **Auditable progressive multimodal bearing fault diagnosis with Large language model Screening.** *Measurement*, 2026.
- Yunuo Liu, Dawei Zhu, Zena Al-Khalili, Dai Cheng, **YanJun Chen**, Dietrich Klakow, Wei Zhang, Xiaoyu Shen. **PricingLogic: Evaluating LLMs Reasoning on Complex Tourism Pricing Tasks.** *EMNLP 2025*.
- Xinghao Chen, Zhijing Sun, Wenjin Guo, Miaoran Zhang, **YanJun Chen**, Yirong Sun, Hui Su, Yijie Pan, et al. **Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning.** *Findings of ACL 2025*.

7. Yirong Sun, Dawei Zhu, **Yanjun Chen**, Erjia Xiao, Xinghao Chen, Xiaoyu Shen. **Fine-Grained and Multi-Dimensional Metrics for Document-Level Machine Translation**. *NAACL 2025 SRW*.
Additional co-authored preprints, 2024 to 2026; full list on *Google Scholar*.

Open Source

C3

github.com/EIT-EAST-Lab/C3

Official implementation of *The Trace Is the State* (arXiv:2603.06859): exact per-decision credit by counterfactual continuation, and evaluation of any credit signal by its bias, variance, and agreement with an independent reference.

AccuracyParadox-RLHF

github.com/EIT-NLP/AccuracyParadox-RLHF

Official implementation of *The Accuracy Paradox in RLHF* (EMNLP 2024): reproducible training pipelines, evaluation harnesses, and reference reward models.

3D-CoT dataset

huggingface.co/datasets/Battam/3D-CoT

Chain-of-thought annotations for 3D vision-language alignment (arXiv:2503.06232).

Myco, independent project

github.com/Battam1111/Myco

Agent infrastructure in Rust: persistent memory and knowledge management for AI agents in the Model Context Protocol (MCP) ecosystem.

Technical Skills

Post-Training	SFT and RLHF; PPO, DPO, GRPO; reward modeling and evaluation; RLVR; SAC and off-policy RL for continuous control.
Training Infra	DeepSpeed (ZeRO), Megatron-LM; verl, OpenRLHF, Hugging Face TRL; LoRA / PEFT; multi-GPU distributed training; vLLM inference.
Frameworks	PyTorch, Hugging Face Transformers / Datasets / Accelerate; Weights & Biases.
Programming	Python, C/C++, Bash; Git, Linux.
Languages	Mandarin (native), English (professional), Japanese (elementary).

Teaching & Service

Teaching (TA)	COMP5221 Software Project Management · COMP1002 Computational Thinking and Problem Solving · COMP1411 Introduction to Computer Systems · COMP5567 Distributed Algorithms and Protocols for Blockchains (2024 to 2026).
Reviewing	ACL Rolling Review (ARR) 2026 · IEEE Transactions on Neural Networks and Learning Systems (TNNLS).